INCH - POUND

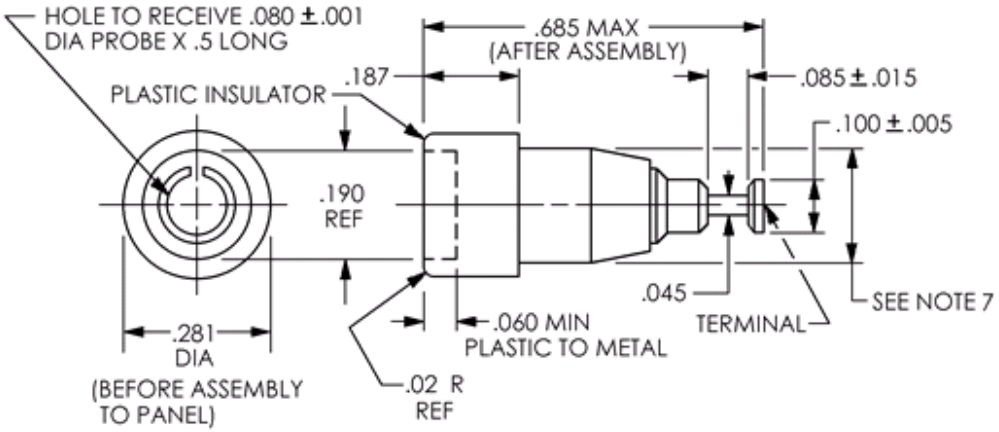
MIL-DTL-39024/14C
w/AMENDMENT 2
3 June 2020
SUPERSEDING
MIL-DTL-39024/14C
w/AMENDMENT 1
01 December 2018

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRICAL, TEST POINT TYPE, PANEL TYPE; SINGLE TEST POINT SUBMINIATURE (PUSH-IN), LOW VOLTAGE, .080

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-39024.



Inches	mm	Inches	mm
.001	0.03	.085	2.16
.005	0.13	.100	2.54
.015	0.38	.187	4.75
.02	0.5	.190	4.83
.045	1.14	.218	5.54
.060	1.52	.281	7.14
.080	2.03	.500	12.70
		.685	17.40

NOTES:
1. Dimensions are in inches. Metric equivalents are given for information only.
2. Unless otherwise specified, tolerances are ± .010 (0.25 mm).
3. Recommended mounting hole diameter .218 ± .001 (5.54 ± 0.03 mm).
4. Range of acceptable panel thickness: .031 - .093 (0.79 – 2.36 mm).
5. Parts supplied unassembled.
6. All undimensioned pictorial configurations are for reference purpose only.
7. When unassembled, this dimension shall be .218 ± .005 (5.54 ± 0.13 mm). When the terminal assembly and insulator are assembled, this dimension shall be .220 (5.59 mm) minimum.

FIGURE 1. Configuration and dimensions.

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TABLE I. Design and identification data.

Dash number	Insulation	
	Material	Color in accordance with SAE-AMS-STD-595
-01	L-P-410 or ASTM D4066	White
-02		Red
-03		Black
-04		Brown
-05		Green
-06		Orange
-07		Blue
-08		Yellow
-09		Gray
-10		Purple
-11		Natural
-12	ASTM D1710	White
-13		Red
-14		Black
-15		Brown
-16		Green
-17		Orange
-18		Blue
-19		Yellow
-20		Gray
-21		Purple
-22		Natural

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1.

Insulation: Material shall be as specified in table I.

Socket contact and terminals: Material shall be as specified in MIL-DTL-39024.

Flammability: Dielectric materials shall be self-extinguishing in accordance with ASTM D635.

Terminals: Shall be capable of being wired with two no. 20 AWG wires. Configuration shall be specified on figure 1.

Test probe: 0.080 ± 0.001 inch in diameter and ½ -inch (min) length. Test probe shall be engaged for a minimum of ½ inch.

Contact current rating: 5 amperes (max).

Contact resistance: A test probe in accordance with MIL-DTL-39024 shall be fully inserted in the connector. With a current of 5 amperes, potential drop shall be measured between the connector terminal and the extreme end of the solder cup on the test probe. Measured values shall be within the limits specified in table II.

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TABLE II. Potential drop (max) with 5 amperes of current.

Test	Before test	After test
Durability	6 mV	8 mV
Vibration	8 mV	10 mV
Salt spray (corrosion)	10 mV	15 mV

Dielectric withstanding voltage (at sea level):

Test voltage: 3,000 volts rms, 60 hertz, shall be applied to the insulting material for period of 15 seconds.

Insertion and withdrawal forces:

Insertion force: 5.0 pounds (max).
Withdrawal force: 0.4 pound (min).

Identification marking:

Part or Identifying Number (PIN): M39024/14-(dash number from table I).

Operating conditions:

Operating voltage: 2,000 volts rms, 60 hertz at sea level; 350 volts rms, 60 hertz at 80,000 feet.

The thickness of the panel on which these connectors may be used ranges from 0.031 to 0.093 inch; the diameter of the mounting hole is 0.218 $\pm$ 0.001 inch.

Qualification: Not applicable.

First article inspection. First article inspection shall consist of Group A and B inspections. MIL-DTL-39024 paragraph 4.5 through 4.5.3.1.3 and Tables III through VII as applicable.

Conformance inspection. Inspection of successive lots for delivery shall consist of Group A inspections. MIL-DTL-39024 paragraph 4.5 through 4.5.3.1.3 and Tables III through VII as applicable.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-39024, this document references the following:

ASTM D635
ASTM D1710
ASTM D4066
SAE-AMS-STD-595
L-P-410

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5935-2020-082)

Review activities:
Army - AR, CR4, MI
Navy - AS, CG, MC, OS
Air Force - 19

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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